



Maxime Lavaud, *PhD*

+(33) 646754510

lavaudmaxime.ml@gmail.com



PROFESSIONAL EXPERIENCE

01/01/2025 - : Postdoctoral Researcher, University of Vienna

- Manage and update the FastDDM software, freely available on Github.
- Microscopy experiments on the effect of hydrodynamic interactions in concentrated sedimenting colloidal suspensions.
- Development of lock-in differential dynamic microscopy, a new technique that measures the zeta potential of colloids, including its sign, with an ordinary microscope.
- Development of holographic optical tweezers to measure colloid–colloid interactions in complex fluids.
- Consultancy for a company on the design of a new space instrument.
- Grant applications (teaching project and proof of concept).

01/01/2024 - 31/12/2024 : Postdoctoral Researcher, CBMN and LOMA, University of Bordeaux.

- Design and implementation of an optical setup** measuring the absorbance of solutions in a microfluidic system, enabling the analysis of Taylor dispersion for the characterization of amyloid protein fibrillation dynamics.
- Assisted in the design of various optical setups (RCIM, TIRF, interferometric microscopy) as well as data analysis (particle tracking, PIV, etc.).
- Team leader of EmetBrown in the AnDi challenge, focused on using deep learning for biomimetic trajectory analysis.
- Code optimization for simulating Brownian motion in complex systems and its adaptation to high-performance computing (HPC).
- Set up and manage the Github repository for the EmetBrown team.

15/12/2022 - 14/12/2023: Research Engineer, EPOC, PROMESS Team, Bordeaux INP.

- Developed digital tools for the systematic analysis of field experiment data, enabling the characterization of interactions between soil and vines.

01/01/2022 - 14/12/2022: Postdoctoral Researcher, LOMA, EmetBrown Team, University of Bordeaux.

- Designed and set up a new experimental laboratory.
- Participated in the design and implementation of the laboratory's microfluidics platform.
- Developed digital tools for the systematic analysis of interferometry videos, enabling the 3D tracking of micrometric particles.
- Optimized code for the simulation of confined Brownian motion and ported it to HPC for large data acquisition, enabling the precise measurement of rare events.

➤ Selected and purchased instruments funded by an ERC grant.

01/09/2018 - 31/12/2021: PhD in Physics, LOMA, University of Bordeaux.

- Experimental study of confined Brownian motion, supervised by Thomas Salez and Yacine Amarouchene.
- Designed and implemented an inverted microscope for 3D tracking of micrometric particles using interferometry.
- Developed digital tools for the analysis of random trajectories to extract surface-particle interactions.

07/01/2020 - 28/02/2020: Visiting Researcher, IPGG, ESPCI, Paris.

- Modified a microscope to integrate the fluorescence recovery after photobleaching (FRAP) method, allowing the study of Brownian motion in hydrogels.

EDUCATION

28/12/2022: Data Scientist Associate certificate, DataCamp.

26/11/2021: PhD in Physics, LOMA, Université de Bordeaux.

- Jury: Thomas Salez, Alois Würger, Lionel Bureau, Christophe Ybert, Yacine Amarouchene, Frédéric Restagno.

25/08/2020: IBM AI Engineering Professional Certificate, Coursera

12/06/2018: Master's Degree in Fundamental Physics and Applications, Université de Bordeaux.

20/08/2016 - 20/07/2017: Erasmus exchange at Stockholm University.

- Participation in the International Physicists' Tournament with the KTH team. Winner of the Swedish national tournament, 8th place internationally.

TEACHING

2026: Differential dynamic microscopy course at a summer school and at the International Soft Matter Conference (ISMC 2026).

2025-2026: Scientific image analysis for Master's and Bachelor's students. Mathematical analysis for 2nd- and 3rd-year Bachelor students. Supervision of five graduate students developing optical experiments and numerical methods.

2024: Supervised two undergraduate students and two graduate students from Lehigh University on exchange. Supervised a postdoc and two third-year undergraduate students on the characterization of PDMS and water inclusions. Supervised a PhD student and a Bachelor student in a numerical challenge.

2022: Trained a PhD student in the experimental and numerical methods developed during my thesis.

2018-2022: Supervised internships, from third-year undergraduate to second-year master's students (4 students).

2018-2022: Preparation for the International Physicists' Tournament (designed experiments and oral presentations).

2018-2020: Laboratory practicals in Mechanics and Magnetostatics.

PUBLICATIONS

- [1] G. Audéoud, L. Moine, L. Bonnetcaze, **M. Lavaud**, L. Khemtemourian, Y. Amarouchene, T. Salez, and M. Mathelié-Guinlet. "A label-free method to quantify early-stage amyloid aggregation under flow via intrinsic phenylalanine fluorescence". In: (Feb. 2026). arXiv: 2602.19817 [cond-mat.soft].
- [2] Q. Thomas, C. M. Sop, **M. Lavaud**, Y. Amarouchene, T. Salez, and P. Damman. "Broadband molecular dynamics simulation of fluid inertial effects in confined Brownian motion". In: (2026). arXiv: 2606.24291 [cond-mat.soft]. URL: <https://arxiv.org/abs/2606.24291>.
- [3] B. Tregon, G. Nique, W. Benharbone, **M. Lavaud**, and A. R. Camboulives. "The blurring interference seeing motion monitor". In: *Research Square* (Feb. 2026).
- [4] J. Lacherez, M. **Lavaud**, Y. Amarouchene, D. S. Dean, and T. Salez. "Enhanced diffusion over a periodic trap by hydrodynamic coupling to an elastic mode". In: *Communication Physics* 8.465 (2025).
- [5] E. Lattuada, F. Krautgasser, **M. Lavaud**, F. Giavazzi, and R. Cerbino. "The Hitchhiker's Guide to Differential Dynamic Microscopy". In: *The Journal of Chemical Physics* 163.16 (Oct. 2025), p. 161501.
- [6] **M. Lavaud**^{*}, Y. Shokeeb, J. Lacherez, Y. Amarouchene, and T. Salez. "Bidirectional mamba state-space model for anomalous diffusion". In: *Journal of Physics: Photonics* (2025).
- [7] N. Fares, **M. Lavaud**, Z. Zhang, A. Jha, Y. Amarouchene, and T. Salez. "Observation of Brownian elastohydrodynamic forces acting on confined soft colloids". In: *Proceedings of the National Academy of Sciences* 121.42 (2024), e2411956121.
- [8] A. Alexandre, **M. Lavaud**[†], N. Fares, E. Millan, Y. Louyer, T. Salez, Y. Amarouchene, T. Guérin, and D. S. Dean. "Non-Gaussian Diffusion Near Surfaces". In: *Physical Review Letters* 130.7 (Feb. 15, 2023), p. 077101. doi: 10.1103/PhysRevLett.130.077101.
- [9] E. Millan, **M. Lavaud**[†], Y. Amarouchene, and T. Salez. "Numerical simulations of confined Brownian-yet-non-Gaussian motion". In: *The European Physical Journal E* 46.4 (Mar. 31, 2023), p. 24. ISSN: 1292-895X.
- [10] **M. Lavaud**, T. Salez, Y. Louyer, and Y. Amarouchene. "Stochastic inference of surface-induced effects using Brownian motion". In: *Physical Review Research* 3.3 (July 8, 2021), p. L032011. doi: 10.1103/PhysRevResearch.3.L032011.
- [11] U. Delabre, N.-A. Goy, A. Grolleau, N. Dufour, **M. Lavaud**, and Z. Denis. "Des travaux pratiques smartphones à l'université". In: *Bulletin de l'Union des Physiciens (1907-2003)* 112.1003 (Apr. 2018).
- [12] N.-A. Goy, Z. Denis, **M. Lavaud**, A. Grolleau, N. Dufour, A. Deblais, and U. Delabre. "Surface tension measurements with a smartphone". In: *The Physics Teacher* 55.8 (Nov. 2017), pp. 498–499. doi: 10.1119/1.5008349.

[†] Co-first authorship * Corresponding Author

CONFERENCES

National: GDR Liquids at Interfaces (Bordeaux 2018, Grenoble 2019), Journée de la Physique Statistique (2019), GDR SLAMM (Biarritz 2021), Vienna Soft Matter Day 2025.

International: APS March Meeting Denver 2020, APS March Meeting 2021, ISMC 2025, Donev meeting 2025, ISMC 2026.

Summer school: Liquids at Interfaces Cargèse 2019, Soft Matter Systems: from fundamentals to Sensory Perception 2026.

SKILLS

Languages: French (native), English (fluent)

Programming Languages: Python, Cython, Matlab, C/C++ (basic), bash (basic)

Data Science: Pandas, Sci-kit Learn, Keras, Pytorch, Tensorflow, Time Series Analysis, Statistical Inference, git, docker, and Kubernetes (basic).

Experimental Science: Design and setup of optical systems: Mie Holography (interferometry), microscopy, fluorescence microscopy, Total Internal Reflection Fluorescence Microscopy (TIRF), Fluorescence Recovery After Photobleaching (FRAP), Rheometry, Atomic Force Microscopy (AFM), Holographic optical tweezers (HOT).

Social Engagement:

Representative of non-permanent staff at LOMA (about 50 people), especially during the Hcéres evaluation.

Student representative at the Physics Doctoral School of the University of Bordeaux (about 300 students).

Representative of non-permanent staff at the College of Science and Technology of the University of Bordeaux (about 10,000 students, about 600 faculty members, and about 100 staff members).

Outreach: 80th anniversary of CNRS (experimental physics demonstration on friction), Long Night of Research, Vienna (experimental physics demonstrations on soft matter).

REFERENCES

Roberto Cerbino, Professor, University of Vienna
roberto.cerbino@univie.ac.at

Thomas Salez, CNRS Research Director (Directeur de recherche), LOMA, University of Bordeaux
thomas.salez@cnrs.fr

Yacine Amarouchene, CNRS Researcher (Chargé de recherche), LOMA, University of Bordeaux
yacine.amarouchene@u-bordeaux.fr

Myriam Schmutz, Full Professor (Professeur des universités), Bordeaux INP
myriam.schmutz@bordeaux-inp.fr